

Spacetime as a Thermodynamic Medium: Operational Constraints on Irreversibility and Gravity

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Abstract

We develop an operational and axiomatic framework for thermodynamic notions in spacetime that does not presuppose any microscopic structure of gravity. Temperature, entropy production, and irreversibility are defined relationally through system–environment response and accessibility, rather than through underlying degrees of freedom. Within this framework, we analyze general consistency constraints on irreversible quantum dynamics coupled to gravity, independent of whether such dynamics are fundamental or effective. We show that Landauer-type bounds act as global bookkeeping constraints on entropy and energy flow, and therefore do not, in general, require local heating; in particular, irreversible dynamics may induce cooling without thermodynamic inconsistency. Simple Lindblad toy models are used to illustrate these points. We discuss implications for objective-collapse phenomenology and for cosmological bookkeeping, emphasizing that these applications are illustrative rather than explanatory. The framework is intended as a conservative foundation for analyzing thermodynamic consistency in gravity–quantum couplings, not as a microscopic theory.

Keywords: thermodynamics, gravity, quantum mechanics, irreversibility, Landauer’s principle, spacetime

1 Introduction

The interplay between gravity, thermodynamics, and quantum theory has repeatedly revealed that macroscopic spacetime physics encodes deep constraints on information, entropy, and irreversibility. Black-hole thermodynamics, Unruh radiation, and

semiclassical horizon physics [1–4] all suggest that spacetime admits thermodynamic descriptions that are remarkably insensitive to microscopic details [5]. At the same time, proposals for non-unitary or irreversible modifications of quantum dynamics [6]—whether fundamental or effective—raise longstanding questions about energy conservation, entropy production, and their compatibility with gravitational dynamics.

A recurring difficulty in this area is that thermodynamic concepts are often invoked in a manner that implicitly assumes microscopic structure: temperature is associated with specific degrees of freedom, entropy with counting of microstates, and irreversibility with explicit dissipation into identifiable baths. In the context of gravity, where microscopic structure remains unknown and may not even be meaningful in the usual sense, such assumptions are problematic. This tension is particularly acute when discussing irreversible quantum dynamics, such as objective-collapse models or effective non-unitary descriptions arising from coarse-graining, and their possible gravitational consequences.

The purpose of this work is to develop a conservative, operational framework for thermodynamic reasoning in spacetime that avoids these pitfalls. We adopt an axiomatic approach in which thermodynamic quantities are defined through response, accessibility, and consistency conditions, rather than through microphysical models. In particular, temperature is treated as an operational parameter characterizing exchange relations between systems, not as a property tied to underlying constituents. Entropy production is defined at the level of irreversible dynamics, without commitment to a specific interpretation of information or measurement.

Within this framework, we revisit the role of Landauer-type bounds in the presence of irreversible quantum dynamics coupled to gravity. Landauer’s principle is often interpreted as implying unavoidable local heating whenever information is erased. We emphasize instead its proper role as a global consistency constraint on entropy and energy bookkeeping. When formulated operationally, Landauer-type bounds constrain the total entropy exported to inaccessible degrees of freedom, but do not, in general, fix the sign or locality of energy flow within a subsystem. As a result, irreversible dynamics may lead to heating or cooling depending on structure, without violating thermodynamic consistency.

To make these points concrete, we analyze simple open-system toy models based on Lindblad dynamics. These models illustrate how non-unitary evolution may be thermodynamically consistent while exhibiting qualitatively different energetic behavior, depending on commutativity and coupling structure. No claim is made that such models describe gravity at a fundamental level; their role is purely illustrative, demonstrating the minimal consistency requirements that any coupling between irreversibility and gravity must satisfy.

Finally, we discuss two classes of applications. First, we comment on implications for objective-collapse phenomenology, where energetic constraints are often invoked as decisive arguments. Second, we explore, at a purely illustrative level, how accumulated irreversibility might enter cosmological bookkeeping, including possible contributions that behave as an effective cosmological term. These discussions are not proposed as explanations of collapse or dark energy, but as consistency checks indicating where operational thermodynamics may impose nontrivial constraints.

Throughout, we deliberately refrain from postulating microscopic spacetime structure, from asserting the fundamentality of collapse, or from offering a solution to the cosmological constant problem. The aim is instead to clarify the conceptual and thermodynamic landscape in which such questions are posed, and to provide a disciplined foundation for future theoretical and experimental analyses of gravity–quantum couplings involving irreversibility.

2 Operational Thermodynamics Without Spacetime Microstructure

To analyze thermodynamic consistency in gravitational settings without assuming microscopic structure, we develop an operational framework based on response relations and accessibility constraints.

2.1 Operational Definition of Temperature

We define temperature operationally through system response rather than through microscopic agitation. A system possesses temperature T if, when weakly coupled to a probe, transition rates satisfy detailed balance at T . This definition requires no assumption about underlying degrees of freedom.

Consider a probe system with energy levels E_i coupled weakly to a target system. The operational temperature T is defined by the requirement that transition rates satisfy:

$$\frac{W(j \rightarrow i)}{W(i \rightarrow j)} = e^{-(E_i - E_j)/(k_B T)} \quad (1)$$

where $W(i \rightarrow j)$ denotes the rate of transitions from state i to state j [7].

This operational approach naturally extends to gravitational contexts. In curved spacetime, local temperature fields $T(x)$ are defined through local detailed balance conditions. For stationary spacetimes, consistency with general covariance implies the Tolman-Ehrenfest relation [8, 9]:

$$T(x) \sqrt{-g_{00}(x)} = \text{const} \quad (2)$$

The key insight from Unruh, Hawking, and de Sitter effects [2–4] is that spacetime exhibits operational temperature when causal structure enforces information inaccessibility. Temperature emerges from horizon-induced restrictions on observability, not from microscopic constituents.

2.2 Entropy Production and Irreversibility

Entropy production is defined at the level of stochastic dynamics, without reference to underlying microstates. For a quantum system evolving under a completely positive, trace-preserving map Φ , the entropy production is:

$$\Delta S = S(\Phi(\rho)) - S(\rho) \geq 0 \quad (3)$$

where $S(\rho) = -\text{Tr}[\rho \log \rho]$ is the von Neumann entropy. More general notions of entropy production for quantum dynamical semigroups are often formulated via relative entropy or Spohn-type inequalities [10]; the present toy discussion does not depend on the specific choice. Here equation 3 should be understood as a schematic indicator of irreversibility; none of the subsequent arguments depend on monotonicity of the von Neumann entropy for arbitrary CPTP maps.

Irreversibility occurs when the map Φ is not invertible on the accessible algebra of observables. This definition applies equally to fundamental non-unitary dynamics and to effective descriptions arising from coarse-graining over inaccessible degrees of freedom.

2.3 Global Versus Local Thermodynamic Constraints

A crucial distinction emerges between global and local applications of thermodynamic principles. Global constraints, such as overall entropy increase, apply to the total system including all coupled sectors. Local constraints apply only to restricted subsystems with specific accessibility properties.

Landauer's principle, properly understood, provides a global constraint on information erasure: the erasure of one bit requires dissipation of at least $k_B T \ln 2$ into some physical sector. This principle does not specify which sector absorbs the entropy, nor does it require that energy flow be positive in every subsystem.

3 Landauer-Type Bounds as Bookkeeping Constraints

3.1 Landauer's Principle Revisited

Landauer's principle [11] states that the erasure of information requires energy dissipation:

$$\Delta E \geq k_B T \ln 2 \times (\text{bits erased}) \quad (4)$$

Applied to irreversible quantum dynamics, this becomes a constraint on total entropy export. If a quantum process reduces the entropy of a subsystem by $\Delta S < 0$, then some other sector must absorb entropy $\Delta S_{\text{env}} \geq -\Delta S$, with corresponding energy cost $\Delta E_{\text{total}} \geq T \Delta S_{\text{env}}$.

3.2 Why Irreversibility Does Not Imply Local Heating

The crucial point is that Landauer's principle constrains the total energy-entropy budget, not the local energy flow in any particular subsystem [12, 13]. Consider a system S coupled to an environment E . If an irreversible process cools the system (reduces its entropy), thermodynamic consistency requires:

1. The environment absorbs the exported entropy: $\Delta S_E \geq -\Delta S_S$
2. Total energy is conserved: $\Delta E_S + \Delta E_E = 0$
3. Total entropy increases: $\Delta S_S + \Delta S_E \geq 0$

These constraints are compatible with $\Delta E_S < 0$ (system cooling) provided the environment satisfies appropriate response relations. The global Landauer bound applies to the environment's entropy absorption, not to local heating requirements.

4 Illustrative Open-System Toy Models

To demonstrate these principles concretely, we analyze simple quantum systems undergoing irreversible evolution [14, 15].

4.1 Thermalizing Lindblad Dynamics

Consider a two-level system with Hamiltonian $H = \frac{\hbar\omega}{2}\sigma_z$ coupled to a thermal environment through Lindblad dynamics:

$$\frac{d\rho}{dt} = \gamma_{\downarrow} \left(\sigma_- \rho \sigma_+ - \frac{1}{2} \{ \sigma_+ \sigma_-, \rho \} \right) + \gamma_{\uparrow} \left(\sigma_+ \rho \sigma_- - \frac{1}{2} \{ \sigma_- \sigma_+, \rho \} \right) \quad (5)$$

Imposing detailed balance at temperature T :

$$\frac{\gamma_{\uparrow}}{\gamma_{\downarrow}} = e^{-\hbar\omega/(k_B T)} \quad (6)$$

This channel thermalizes the system to the Gibbs state $\rho_{\text{eq}} = e^{-H/(k_B T)} / \text{Tr}[e^{-H/(k_B T)}]$ while maintaining thermodynamic consistency.

4.2 Cooling Under Irreversible Evolution

Not all irreversible dynamics lead to heating. Consider a modified Lindblad generator where the rates violate detailed balance:

$$\frac{\gamma_{\uparrow}}{\gamma_{\downarrow}} = \alpha e^{-\hbar\omega/(k_B T)} \quad (7)$$

with $\alpha < 1$.

This process preferentially induces downward transitions, cooling the system below the environmental temperature. Thermodynamic consistency is maintained if the excess entropy is exported to an inaccessible sector, satisfying the global Landauer constraint. This is a non-equilibrium irreversible channel, but we once again emphasize that Landauer consistency is global.

4.3 Commutativity and Energetic Structure

The energetic consequences of irreversible dynamics depend critically on the commutation relations between the system Hamiltonian and the Lindblad operators. When $[H, L_i] = 0$ for all collapse operators L_i , energy is conserved on average during each collapse event. When the operators do not commute, energy exchange occurs.

For cooling to be thermodynamically consistent, the collapse operators must satisfy:

1. Non-unital structure: $\sum_i L_i^\dagger L_i \neq \mathbb{I}$
2. Energetic bias: Average energy change per collapse event is negative
3. Entropy export: Reduced system entropy is compensated by environmental entropy increase

5 Illustrative Applications

5.1 Implications for Objective-Collapse Phenomenology

Objective-collapse models [6] propose that wavefunction reduction is a fundamental physical process. If collapse involves information erasure, Landauer's principle applies, requiring energy dissipation into some sector of nature.

Our framework clarifies that this energy dissipation need not manifest as local heating of the collapsing system. Depending on the structure of the collapse operators, the process may induce cooling while exporting entropy to gravitational or other inaccessible degrees of freedom [16, 17].

This observation is relevant for experimental tests of collapse models, which often look for excess heating as a signature of objective reduction. Our analysis shows that cooling signatures may be equally consistent with collapse phenomenology, depending on the details of the underlying dynamics.

5.2 Cosmological Bookkeeping and an Effective Cosmological Term

At a purely illustrative level, we can explore how accumulated irreversibility might contribute to cosmological dynamics [18]. If irreversible processes systematically export entropy to gravitational degrees of freedom, the accumulated energy-entropy exchange might contribute an effective stress-energy component.

Consider a collection of irreversible processes throughout cosmic history, each exporting entropy ΔS_i to gravitational degrees of freedom. The total accumulated entropy export is:

$$S_{\text{total}} = \sum_i \Delta S_i \quad (8)$$

If this entropy is associated with an effective energy scale T_{eff} , the corresponding energy density contribution is:

$$\rho_{\text{eff}} \sim T_{\text{eff}} S_{\text{total}} / V_{\text{universe}} \quad (9)$$

For sufficiently small T_{eff} and large accumulated entropy, this could contribute a small, positive energy density with equation of state resembling a cosmological constant.

This speculation is offered purely as a consistency check, not as an explanation of dark energy. It illustrates how operational thermodynamics might constrain cosmological phenomenology without requiring fundamental modifications of general relativity. No claim is made that the effective temperature T_{eff} corresponds to physical bath or

horizon temperature. Also, and in particular, no dynamic coupling between collapse events and the metric is assumed or derived here.

6 Scope and Limitations

The framework developed in this work is intentionally conservative in scope. It is designed to clarify thermodynamic and informational consistency conditions on couplings between irreversible quantum dynamics and gravity, rather than to propose a microscopic theory or a specific mechanism.

In particular, we emphasize the following limitations:

- We do not postulate any microscopic structure of spacetime, nor do we assume the existence of spacetime degrees of freedom beyond those implicitly required by semiclassical gravity.
- We do not assume that non-unitary or irreversible quantum dynamics are fundamental; the analysis applies equally to effective descriptions arising from coarse-graining or environmental inaccessibility.
- We do not claim to resolve the cosmological constant problem or to provide a dynamical explanation of dark energy. Cosmological considerations are presented solely as illustrative consistency checks.
- We do not assert that Landauer’s principle mandates local heating or dissipation in any specific subsystem. All uses of Landauer-type bounds are global and bookkeeping in character.
- We do not propose a specific experimental realization in this work; experimental considerations are discussed only to indicate potential discriminants and consistency constraints.

Accordingly, the results presented here should be understood as conceptual and structural. Their value lies in clarifying which classes of assumptions are thermodynamically admissible, and which are not, when discussing irreversible quantum dynamics in gravitational settings.

7 Discussion and Outlook

The operational framework developed here highlights a recurring theme in the foundations of gravity and quantum theory: many apparent paradoxes arise not from dynamics, but from unexamined assumptions about thermodynamic concepts. By defining temperature, entropy production, and irreversibility relationally rather than microscopically, one can formulate nontrivial consistency constraints without committing to speculative spacetime ontology.

Several directions for further work suggest themselves. On the theoretical side, it would be valuable to explore whether the effective negative-pressure behavior discussed in illustrative cosmological bookkeeping can be derived more systematically from coarse-grained stress–energy considerations, such as bulk-viscous or constraint-based formulations. On the experimental side, precision measurements of heating,

cooling, and fluctuation–dissipation balance in mesoscopic quantum systems may provide indirect probes of classes of irreversible dynamics consistent with the present framework.

More broadly, the analysis underscores the utility of thermodynamics as a structural guide in regimes where microscopic descriptions remain unavailable. Whether gravity ultimately admits a quantum description that is strictly unitary, effectively non-unitary, or something else entirely, any such theory must remain compatible with the operational thermodynamic constraints articulated here.

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